

which is expected to go down in the coming years. Green hydrogen holds potential for utilisation in both the blast furnace process and in the direct reduced iron (DRI) route of net-zero steel production. But steelmakers can use natural gas until green hydrogen becomes commercially viable. Also, we can explore carbon capture, utilisation and storage (CCUS) technologies, among others, that are being developed.

The Role of Government and Policies

The government should adopt a carrot-and-stick policy to accelerate green transition. Also, it needs to implement uniform measurement and reporting standards to measure carbon footprint. Demand for net-zero steel from the government can also boost its production. A

centives (for three years) are a step in the right direction; but they need to be increased both in terms of volume and duration. The government's critical role also lies in ensuring that developed nations provide both technological and financial support to the developing countries for achieving net-zero emissions.

A Synergistic Path

Supply chain management is fundamental for sustainable practices at every stage. For instance, steel is a key raw material in the manufacturing of RE elements, including solar panels, wind turbines, and transmission towers. In the manufacturing of vehicles, significant benefits derive from using low-carbon steel that is stronger and lighter too. A car



ILLUSTRATION BY RAJ VERMA

favourable tax structure will encourage capital investment, while bold policies can create new markets. Net-zero steel users should also be incentivised.

Implementation of policies to facilitate the reduction of renewable power costs, particularly by allowing for an increased capacity of solar panels, should be a high priority. Setting aside the 40 per cent customs duty on solar panels will help enhance solar panel manufacturing. Also, natural gas should be brought under GST. As the net-zero goal leads to increased use of scrap metal, it is imperative to optimise domestic availability. While it is necessary to ramp up transmission capacities, the waiver of transmission charges should be extended for an additional two years beyond June 2025. The recently launched green hydrogen production in-

that weighs less enhances fuel efficiency, resulting in lower energy consumption and emissions.

Apart from this, collaboration between larger firms and start-ups can provide a win-win situation, inspiring disruptive ideas and technologies. Overall, we need a multi-faceted approach involving government policies, industry actions, start-up innovation, and collaboration to transition India away from fossil fuels. And the G20 presidency presents India with an opportunity to effectively steer the grouping towards a brighter future with its potential to champion sustainable practices, advocate equitable solutions and foster international collaboration. **BT**

Views are personal